# AUGUSTIN MUYL

+1 (786) 616-2355 | muyl.augustin@gmail.com | github.com/augustinmuyl | linkedin.com/in/augustinmuyl

## Education

**University of Michigan** Ann Arbor, MI
*Bachelor of Science, Honors Mathematics, Computer Science* *May 2028*

- **Relevant Coursework:** Data Structures & Algorithms, Discrete Mathematics, Linear Algebra I and II, Algebraic Structures, Multivariate Calculus, Differential Equations, Electricity & Magnetism, Mechanics & Thermodynamics

## Skills

- **Programming Languages**: Python (FastAPI, Flask), Java, TypeScript (React/Next, Node), HTML/CSS, SQL, Bash
- **Technologies**: NumPy, Pandas, scikit-learn, PyTorch, Snowflake, PostgreSQL, MongoDB, Dataiku, Git, UNIX, Docker

## Experience

**Machine Learning Intern** | *Detect Inspections* May 2026 – Aug 2026

- Built and deployed a damper detector for aerial power-line inspection imagery, achieving 0.99 F1 on a held-out test set, by creating and annotating a 5,250+ box dataset and training a YOLO11s model with custom containment-based NMS
- Surfaced 75+ field-confirmed slid and missing dampers, invisible to standard classifiers, across ~1,000 drone inspection frames by designing a rotation-invariant geometric representation of power-line structures layered on detector outputs
- Re-engineered the embedding readout strategy of a DINOv3 component re-identification pipeline, raising cross-viewpoint matching accuracy up to 40% (0.63→0.88 AUC), validated on a 2,500+ pair benchmarking harness

**Data Science Intern** | *CMA CGM - Group Security & Intelligence* May 2025 – Aug 2025

- Engineered scalable ETL pipelines in Dataiku and Snowflake, optimizing SQL performance to cut processing time of 800M+ container logs from ~20 hours to under 1 hour and enabling near-real-time analysis of high-risk containers
- Developed full-stack features within an internal web application (React, FastAPI), building multiple frontend components and backend APIs used daily by 50+ analysts across 5 continents to streamline investigative workflows
- Designed an algorithm using H3 spatial indexing to identify shippers' likely origin zones with 75%+ accuracy by filtering out hubs/ports and reconstructing average routes, enabling anomaly detection across global shipping patterns

**Undergraduate Researcher** | *Miami Dade College* Sep 2025 – May 2026

- Built a Python simulation framework to study Laplacian spectral robustness across multiple graph families and attack strategies (random, degree, betweenness, Fiedler), with a CLI for graph parameters and removal checkpoints
- Identified that Kirchhoff index strongly predicts GCC size under random node removal ($r = 0.98$) but degrades under targeted degree attacks ($r = 0.76$), suggesting spectral robustness proxies must be selected relative to the threat model

## Projects

**MLP-NumPy** | [github] Jul 2025

- Implemented a Multilayer Perceptron from scratch in NumPy, deriving backpropagation equations for sigmoid activation and binary cross-entropy loss, and designing a modular architecture supporting variable hidden layers
- Built a CLI with configurable training parameters (architecture, learning rate, early stopping) and visualization tools (loss curves, predictions, decision boundaries), to enable experimentation
- Achieved 98% accuracy on MNIST and 89% on Fashion-MNIST, extended to CIFAR-10 with ~48% accuracy

**Vocatio** | [website] [github] Oct 2024 – Feb 2026

- Built and led a full-stack civic-tech platform using Next.js, FastAPI, SQLModel, and PostgreSQL to match students with nonprofit volunteer opportunities based on interests, availability, and location/theme preferences
- Designed and shipped scalable backend services, including search/filter/pagination APIs, slug-based routing, validated CSV bulk-import tooling, and waitlist onboarding with bot protection, deduplication, and async confirmation emails
- Selected for the First-Year Innovation Fellowship at Innovate@BU for the project's civic impact, receiving $1,000 in funding, mentorship, and institutional support to expand its reach

## Languages

- French (Native Proficiency), Spanish (Native Proficiency), English (Full Bilingual Proficiency)